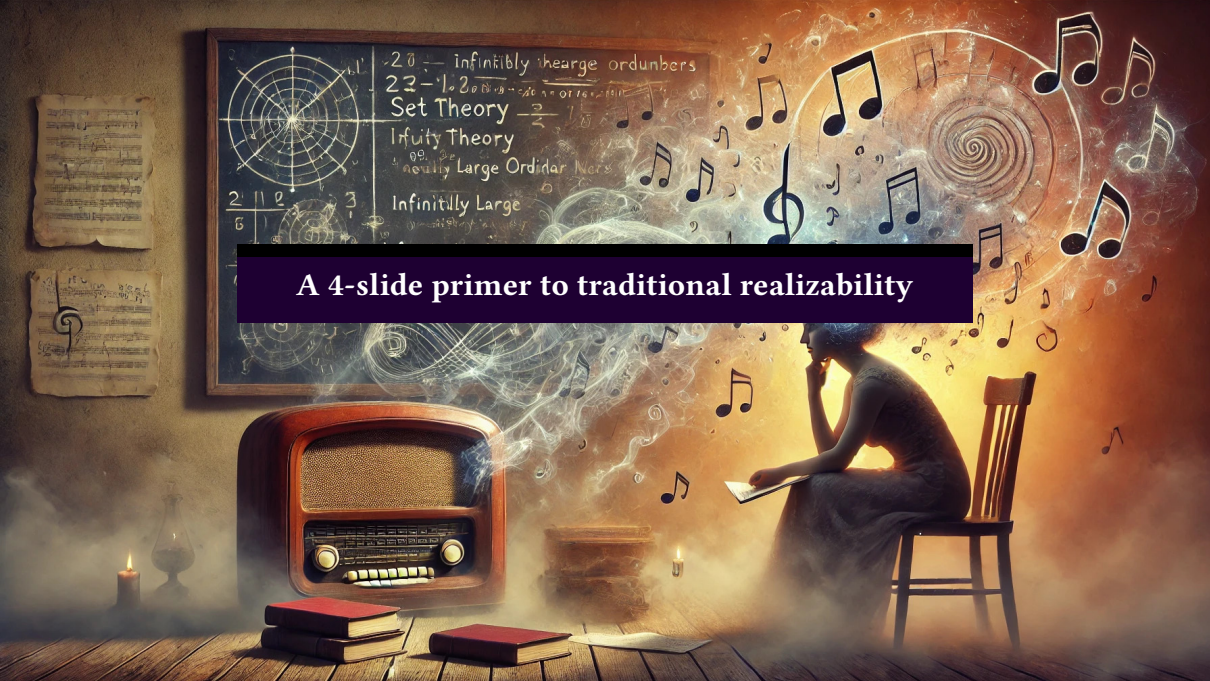

A 4-slide primer to traditional realizability

Thm. For every number $n \in \mathbb{N}$, there is a prime larger than n.

Proof. Any prime factor of $n! + 1$ will do.

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- Integrated developments: *SAT checking, …*
- Computability theory: *induction* $\widehat{=}$ *recursion, …*
- Metatheory of constructive systems: *provability results, …*
- Philosophy of proof and computation: *realizability in the real world, …*

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- Integrated developments: *SAT checking, …*
- Computability theory: *induction* $\widehat{=}$ *recursion, …*
- Metatheory of constructive systems: *provability results, …*
- Philosophy of proof and computation: *realizability in the real world, …*

Thm. Every function $\alpha : \mathbb{N} \to \mathbb{N}$ is *good* in that there are $i < j$ such that $\alpha(i) \leq \alpha(j)$.

Proof. By LEM, there is a minimal value $\alpha(i)$. Set $j := i + 1$.

"**Every** theorem has a **computable*** **witness**."

* with monadic side effects

Number realizability

$e \Vdash s = t$	iff $s = t$.
$e \Vdash \top$	iff true.
$e \Vdash \bot$	iff false.
$e \Vdash (\varphi \wedge \psi)$	iff $\pi_1 \cdot e \downarrow$ and $\pi_2 \cdot e \downarrow$ and $\pi_1 \cdot e \Vdash \varphi$ and $\pi_2 \cdot e \Vdash \psi$.
$e \Vdash (\varphi \vee \psi)$	iff $\pi_1 \cdot e \downarrow$ and $\pi_2 \cdot e \downarrow$ and if $\pi_1 \cdot e = 0$ then $\pi_2 \cdot e \Vdash \varphi$, and if $\pi_1 \cdot e \neq 0$ then $\pi_2 \cdot e \Vdash \psi$.
$e \Vdash (\varphi \Rightarrow \psi)$	iff for every $r \in \mathbb{N}$ such that $r \Vdash \varphi$, $e \cdot r \downarrow$ and $e \cdot r \Vdash \psi$.
$e \Vdash (\forall n : N.\, \varphi(n))$	iff for every $n_0 \in \mathbb{N}$, $e \cdot n_0 \downarrow$ and $e \cdot n_0 \Vdash \varphi(n_0)$.
$e \Vdash (\exists n : N.\, \varphi(n))$	iff $\pi_1 \cdot e \downarrow$ and $\pi_2 \cdot e \downarrow$ and $\pi_2 \cdot e \Vdash \varphi(\pi_1 \cdot e)$.
$e \Vdash (\forall f : N^N.\, \varphi(f))$	iff for every $f_0 : \mathbb{N} \to \mathbb{N}$ and every $r_0 \in \mathbb{N}$ such that f_0 is computed by the r_0-th machine, $e \cdot r_0 \downarrow$ and $e \cdot r_0 \Vdash \varphi(f_0)$.
$e \Vdash (\exists f : N^N.\, \varphi(f))$	iff $\pi_1 \cdot e \downarrow$ and $\pi_2 \cdot e \downarrow$ and the $(\pi_1 \cdot e)$-th machine computes a function $f_0 : \mathbb{N} \to \mathbb{N}$ and $\pi_2 \cdot e \Vdash \varphi(f_0)$.

Thm. If HA $\vdash \varphi$, then there is a number $e \in \mathbb{N}$ such that $e \Vdash \varphi$.

Number realizability

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$e \Vdash (\forall n : N.\, \varphi(n))$	iff for every $n_0 \in \mathbb{N}$, $e \cdot n_0 \downarrow$ and $e \cdot n_0 \Vdash \varphi(n_0)$.
$e \Vdash (\exists n : N.\, \varphi(n))$	iff $\pi_1 \cdot e \downarrow$ and $\pi_2 \cdot e \downarrow$ and $\pi_2 \cdot e \Vdash \varphi(\pi_1 \cdot e)$.
$e \Vdash (\forall f : N^N.\, \varphi(f))$	iff for every $f_0 : \mathbb{N} \to \mathbb{N}$ and every $r_0 \in \mathbb{N}$ such that f_0 is computed by the r_0-th machine, $e \cdot r_0 \downarrow$ and $e \cdot r_0 \Vdash \varphi(f_0)$.
$e \Vdash (\exists f : N^N.\, \varphi(f))$	iff $\pi_1 \cdot e \downarrow$ and $\pi_2 \cdot e \downarrow$ and the $(\pi_1 \cdot e)$-th machine computes a function $f_0 : \mathbb{N} \to \mathbb{N}$ and $\pi_2 \cdot e \Vdash \varphi(f_0)$.

Thm. If HA $\vdash \varphi$, then there is a number $e \in \mathbb{N}$ such that HA $\vdash (\underline{e} \Vdash \varphi)$.

The effective topos

	statement	in Set	in Eff(TM)	in Eff(ITTM)
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	?	?
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	?	?
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

The effective topos

	statement	in Set	in $\mathrm{Eff}(\mathrm{TM})$	in $\mathrm{Eff}(\mathrm{ITTM})$
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	?	?
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	?	?
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

"$\mathrm{Eff}(\mathrm{TM}) \models$ 1" amounts to: There is a Turing machine which determines of any given number whether it is prime or not.

The effective topos

	statement	in Set	in $\mathrm{Eff}(\mathrm{TM})$	in $\mathrm{Eff}(\mathrm{ITTM})$
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	?	?
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	?	?
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

"$\mathrm{Eff}(\mathrm{TM}) \models$ 2" amounts to: There is a Turing machine which, given a number n, computes a prime larger than n.

The effective topos

	statement	in Set	in $\mathrm{Eff}(\mathrm{TM})$	in $\mathrm{Eff}(\mathrm{ITTM})$
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	?	?
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	?	?
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

"$\mathrm{Eff}(\mathrm{TM}) \models$ 3" amounts to: There is a Turing machine which, given a Turing machine computing a map $f : \mathbb{N} \to \mathbb{N}$, determines whether f has a zero or not.

The effective topos

	statement	in Set	in $\mathrm{Eff}(\mathrm{TM})$	in $\mathrm{Eff}(\mathrm{ITTM})$
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	?
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	?	?
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

"$\mathrm{Eff}(\mathrm{TM}) \models$ 4" amounts to: There is a Turing machine which, given a Turing machine computing a map $f : \mathbb{N} \to \mathbb{N}$, outputs a Turing machine computing f.

The effective topos

	statement	in Set	in Eff(TM)	in Eff(ITTM)
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	✗
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	?	?
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

"Eff(ITTM) $\models$ 4" amounts to: There is an infinite time Turing machine which, given an *infinite time* Turing machine computing a map $f : \mathbb{N} \to \mathbb{N}$, outputs an (*ordinary*) Turing machine computing f.

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statement	in Set	in Eff(TM)	in Eff(ITTM)
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2 Beyond every number there is a prime.	✓	✓	✓
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4 Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	✗
5 Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	✓ (if MP)	?
6 Markov's principle holds.	✓ (trivially)	?	?
7 Countable choice holds.	✓	?	?
8 There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

A real number of Eff(TM) is externally represented by a Turing machine M which on input n outputs a rational approximation $M(n)$. These approximations need to be compatible in that $|M(n) - M(m)| \leq 2^{-n} + 2^{-m}$ for all n, m.

Two such machines M and M' represent the same real number iff $|M(n) - M'(m)| \leq 2^{-n} + 2^{-m}$ for all n, m.

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3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	✗
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	✓ (if MP)	✗
6	Markov's principle holds.	✓ (trivially)	?	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

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5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	✓ (if MP)	✗
6	Markov's principle holds.	✓ (trivially)	✓ (if MP)	?
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

Markov's principle states: $\forall f : \mathbb{N} \to \mathbb{N}. \neg\neg(\exists n \in \mathbb{N}. f(n) = 0) \Rightarrow (\exists n \in \mathbb{N}. f(n) = 0)$.

"Eff(TM) $\models$ 6" amounts to: There is a Turing machine which, given a Turing machine computing a map $f : \mathbb{N} \to \mathbb{N}$ and given the promise that it is *not not* the case that f has a zero, determines a zero of f.

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	statement	in Set	in Eff(TM)	in Eff(ITTM)
1	Every number is prime or not prime.	✓ (trivially)	✓	✓
2	Beyond every number there is a prime.	✓	✓	✓
3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	✗
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	✓ (if MP)	✗
6	Markov's principle holds.	✓ (trivially)	✓ (if MP)	✓ (if MP)
7	Countable choice holds.	✓	?	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

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3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	✗
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	✓ (if MP)	✗
6	Markov's principle holds.	✓ (trivially)	✓ (if MP)	✓ (if MP)
7	Countable choice holds.	✓	✓ (always!)	?
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	?	?

Countable choice states: $\big(\forall x \in \mathbb{N}.\, \exists y \in A.\, \varphi(x, y)\big) \Rightarrow \big(\exists f : \mathbb{N} \to A.\, \forall x \in \mathbb{N}.\, \varphi(x, f(x))\big)$.

"Eff(TM) $\models$ 7" amounts to: There is a Turing machine which, given a Turing machine computing for every $x \in \mathbb{N}$ some $y \in A$ together with a witness of $\varphi(x, y)$, outputs a Turing machine computing a suitable choice function $\mathbb{N} \to A$.

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3	Every map $\mathbb{N} \to \mathbb{N}$ has a zero or not.	✓ (trivially)	✗	✓
4	Every map $\mathbb{N} \to \mathbb{N}$ is computable.	✗	✓ (trivially)	✗
5	Every map $\mathbb{R} \to \mathbb{R}$ is continuous.	✗	✓ (if MP)	✗
6	Markov's principle holds.	✓ (trivially)	✓ (if MP)	✓ (if MP)
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8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	✗	?

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7	Countable choice holds.	✓	✓ (always!)	✓ (always!)
8	There is an injection $\mathbb{R} \hookrightarrow \mathbb{N}$.	✗	✗	✓

Curious size phenomena

$\mathrm{Eff}(\mathrm{ITTM}) \models$ "There exists an injection $\mathbb{R} \hookrightarrow \mathbb{N}$."

means:

There is an infinite time Turing machine which inputs the source of an infinite time Turing machine A representing a real number and outputs a natural number $n(A)$ such that $n(A) = n(B)$ if and only if A and B represent the same real.

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This statement is witnessed by following infinite time Turing machine:

> Read the source of an infinite time Turing machine A from the tape. Simulate all infinite time Turing machines in a dovetailing fashion. As soon as a machine is found which represents the same real as A, output the index of this machine and halt.